

# The PING rate is set by the per-synapse weight, not the pool size

exp047 · 8 June 2026

## Abstract

The PING rate is a function of the per-synapse inhibitory weight  $W^{IE}$  (measured against a fixed release level  $g_i^*$ ), **not** of the pool size  $N_I$ . Sweeping the two independently, the rate-vs- $W^{IE}$  curves for every  $N_I$  collapse onto one.

## Methods

**1. The mechanism, in words.** In a PING cycle the E volley recruits I, the I spikes shunt each E cell toward the inhibitory reversal  $E_i$ , and E stays quiet until that shunt decays enough for the drive to climb  $V$  back to threshold. The cycle’s clock, and hence the rate, is that recovery: how far one inhibitory event drives the conductance past the level that silences E, and how long it takes to decay back. This is a *per-event* quantity. Adding more I cells delivers the same shunt through more synapses; it does not change how deep a single event pins a cell or how long the pin lasts. So the rate should track the per-synapse strength, not  $N_I$ . The rest of this section makes that precise.

**2. The invariant release level.** Reduce to one representative E cell, the COBANet membrane of ar003, with the feedforward excitation written as a tonic current  $I_{\text{ext}}$  as in exp033. This single-cell picture is a heuristic to *motivate* the prediction below, not a proof of it; the full network (§5, Figure 1) is what tests it. When the inhibitory conductance varies slowly the voltage tracks its instantaneous steady state  $V_\infty$  (ar003, eqs 11–13; valid in the high-conductance state of a volley, where the open conductances pull the effective membrane time constant below  $\tau_{\text{GABA}}$ ). Setting  $V_\infty = V_{\text{th}}$  gives the conductance at which the cell is *released* to fire again,

$$g_i^* = \frac{I_{\text{ext}} - I_{\text{th}}}{V_{\text{th}} - E_i}, \quad I_{\text{th}} \equiv g_L(V_{\text{th}} - E_L). \quad (1)$$

with  $g_L$  the leak conductance and  $E_L$  the leak reversal (ar003’s membrane constants),  $V_{\text{th}}$  the spike threshold, and  $E_i$  the inhibitory reversal. So  $g_i^*$  is built from intrinsic constants and the drive alone: **it contains neither  $N_I$  nor the per-synapse weight  $W^{IE}$** . It is the level the cycle is pinned to.

**3. The cycle clock.** Each inhibitory spike adds  $W^{IE}$  to  $g_i$ , which then decays with  $\tau_{\text{GABA}}$  (ar003’s exponential synapse, eq 8). After a volley peaks at  $g_i^{\text{pk}}$  (the sum of the fresh kicks), the conductance decays back through  $g_i^*$  after

$$T_{\text{sup}} = \tau_{\text{GABA}} \ln \frac{g_i^{\text{pk}}}{g_i^*}, \quad (2)$$

so the gamma period is  $T \approx T_{\text{build}} + T_{\text{sup}}$  and  $r_E \propto 1/T$ . A self-organized loop recruits I only until  $g_i$  crosses  $g_i^*$  and E shuts off, so the volley overshoots by about one quantum,  $g_i^{\text{pk}} \approx g_i^* + W^{IE}$ , and

$$T_{\text{sup}} \approx \tau_{\text{GABA}} \ln \left( 1 + \frac{W^{IE}}{g_i^*} \right). \quad (3)$$

**4. The prediction: collapse onto  $W^{IE}$ .** Equation (3) has no  $N_I$  in it. The suppression, hence the rate, is a function of the per-event jump  $W^{IE}$  relative to the invariant  $g_i^*$  alone,

increasing (rate falling) with  $W^{IE}$  and saturating logarithmically. The test is therefore to vary  $W^{IE}$  and  $N_I$  *independently* and plot the rate against  $W^{IE}$ : if (3) holds, the curves for different  $N_I$  must lie on top of one another.

**5. Simulation.** Untrained PING at  $N_E = 1024$ ,  $\Delta t = 0.1$  ms,  $T = 200$  ms. Per-synapse  $W^{IE}$  swept over  $\{0.5, 1, 2, 4, 8, 16\}$   $\mu\text{S}$  (spread held at 10% of the mean) and pool size  $N_I$  over  $\{16, 64, 256\}$ , the two varied on an independent grid. Fixed  $W^{EI} \sim \mathcal{N}(1.0, 0.1)$   $\mu\text{S}$  and uniform 25 Hz Poisson input; inference only. The biophysical default is  $W^{IE} = 2$   $\mu\text{S}$ .

## Results

The curves collapse (Figure 1). At each per-synapse weight the three pool sizes agree to within  $\approx 0.1$  Hz (at the default  $W^{IE} = 2$   $\mu\text{S}$ ,  $r_E = 6.42, 6.44, 6.45$  Hz for  $N_I = 16, 64, 256$ ), and the rate falls smoothly with  $W^{IE}$ , from  $\approx 9.4$  Hz at 0.5  $\mu\text{S}$  to  $\approx 4.2$  Hz at 16  $\mu\text{S}$ , the decelerating, saturating dependence eq (3) predicts (the precise functional form is not pinned by six points, and turning the cycle period into a per-cell rate would need a participation factor (3) does not supply). The I rates (right panel) collapse the same way. Pool size does not move the rate; the per-synapse weight does.

The lesson for sweeping or training  $N_I$ : the rate is governed by the per-synapse  $W^{IE}$  measured against  $g_i^*$ . Hold  $W^{IE}$  fixed and the rate is invariant to pool size; change  $N_I$  alone, at fixed per-synapse weight, and nothing moves.

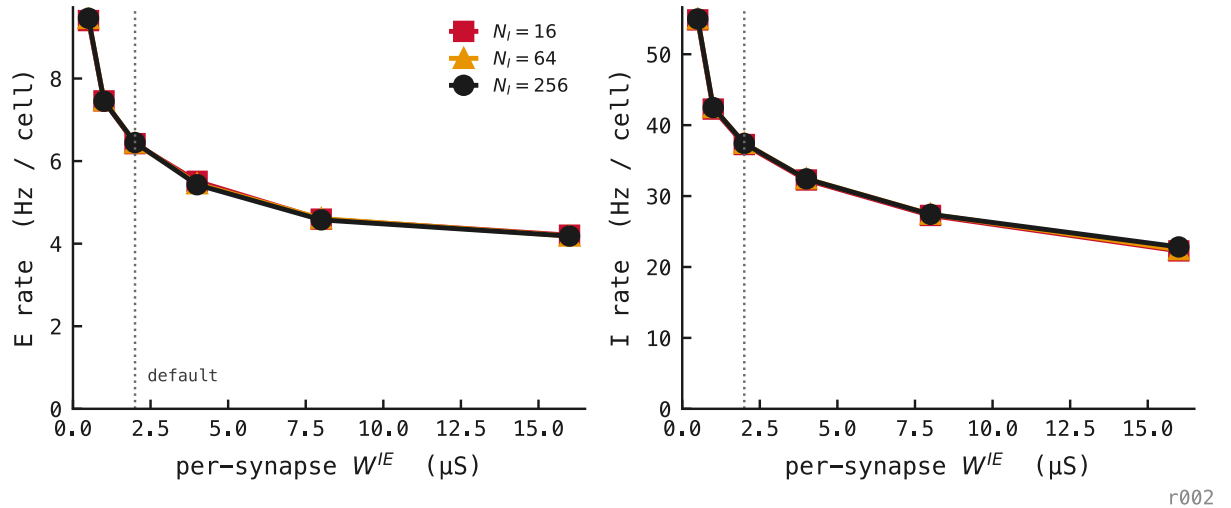


Figure 1: E (left) and I (right) per-cell rate versus the per-synapse weight  $W^{IE}$ , one line per pool size  $N_I$ . The lines collapse: at any fixed  $W^{IE}$  the rate is the same for  $N_I = 16, 64$ , or  $256$ , while it falls steadily with  $W^{IE}$ , the logarithmic, saturating dependence of equation (3). The rate is set by the per-synapse kick, not the pool size. Dotted line marks the biophysical default  $W^{IE} = 2$   $\mu\text{S}$ .